

TEXDraw

Manual

TEX-typing plugin for Unity

Version 5.0.0
2019 August

1 Introduction to TEX

TEXDraw brings the creation of equations and other beauties of TEX-typing in Unity. For those who don't know what's TEX: TEX is a typesetting system mostly designed by **Donald E. Knuth** and first released around 1987.

At the time it became popular in the eye of academia: from scientist to mathematician, because how easily TeX can produce high quality book with minimal effort. TeX is then heavily improved with other derivation works such as LaTeX and ConTeXt. Until today it's still used by thousand TeX enthusiasts around the world.

TeX was rising primilary because it can generate high quality mathematics equations with little effort, for instance:

$$\frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

That can be generated from:

```
$$-b \pm \sqrt{b^2 - 4ac} \over 2a$$
```

Another benefit from using TeX is that it's packaged with over thousands definitions of new symbols and commands, such that it enables you to:

1. Use accents and special characters, e.g.:

- senñorita from `sen\~orita`
- mīnūs from `m\={\i}n\u{u}s`
- Æsop’s Œuvres en français from
`{\AE}sop’s {\OE}uvres en fran\c{c}ais`

2. Use greek letters, e.g.:

- Θ , Φ , Ψ from `$\Theta$`, `$\Phi$`, `$\Psi$`
- α , β , γ from `$\alpha$`, `$\beta$`, `$\gamma$`

3. Use font ligatures, e.g.:

- affirmative from `affirmative`
- en–dash and em—dash from `en--dash` and `em---dash`
- “smart quotes” from `‘‘smart quotes’’`

4. Use font variants, e.g.:

- *Slanted* text
- *Italic* text
- **Boldface** text
- **Typewriter** text
- **Sans-serif** text
- **SMALLCASE** text
- ***Any combination of above.***

You can discover more from separate `reference.tex` file or other T_EX sources like The TeXBook written by Knuth itself. Also take a note: TeX has been developed for more than thirty years in form of its successor, like L^AT_EX— So there’s much to discover more to learn about TeX.

2 What *TEXDraw* Offers

TEXDraw is a plugin to let you type T_EX in Unity. It displays text from interpreting your T_EX either directly or reading from file (such as Resources

folder). *TEXDraw* aims to output 1:1 with plain $\text{T}_{\text{E}}\text{X}$ or $\text{\LaTeX}$ so it also integrates well with other services using $\text{T}_{\text{E}}\text{X}$ technology.

One of reasons you need *TEXDraw* in your Unity project is:

1. You need to write math equations in Unity.
2. You need to display documentations written in $\text{T}_{\text{E}}\text{X}$.
3. You need a component that handles static long-winded text reliably
4. You need one of other features that $\text{T}_{\text{E}}\text{X}$ or *TEXDraw* only offers (tabular text, RTL mode, etc.)

Starting to write in *TEXDraw* is easy: You just have to create a new *TEXDraw* Gameobject with is available via **Create** -> **UI** -> **TEXDraw** in Hierarchy dropdown, then start typing $\text{T}_{\text{E}}\text{X}$ inside of *TEXDraw* component in inspector.

3 What *TEXDraw* Differs

TEXDraw aims to close with $\text{T}_{\text{E}}\text{X}$ but it will never be the same, because major difference in backend technology. Most of differences factor in the fact that $\text{T}_{\text{E}}\text{X}$ is a statistically compiled typesetting, but *TEXDraw* is designed to be dynamically interpreted.

This means *TEXDraw* can render most of the TeX features in small fraction of second without any preliminary checking — and it's good since you want your project keep to running smoothly during initial rendering. Also, unlike other compilers, *TEXDraw* will not complain when there's anything wrong with your TeX file so you can see the change in instant anytime you change your TeX file.

But as compiled vs. interpreted wars going on, there's some features in $\text{T}_{\text{E}}\text{X}$ that is not getting into *TEXDraw* as it's either impossible or would waste a good amount of render time:

- You can't extend *TEXDraw* with `\usepackage`.
- *TEXDraw* only renders to Mesh buffers, hence can't render to external files like `.DVI` or `.PDF`.

- TEXDraw can't autodetect hyphenation and “badness” out of the box. (FYI: Hyphenation is a good feature to “sp-lit wo-rd in-to syl-la-bels” so your document can break wraps in syllables if necessary, hence your document will still look good)
- You can't create new variables or environments with `\def` or `\newenvironment` — You have to do it on either at code-level or project-wide configurations. This is also true for logic gates `\ifx`, macros, value-tokens, etc.
- T_EX gives you unlimited choices for choosing which font variations to suit your documents. *TEXDraw* is not, and it's capped at 31 font variations for one project. (and all of them already been used by built-in package (although you can remove some of them anyway))

4 Changing the Default Stuff in *TEXDraw*

In every TEXDraw GameObject you can change overall document size in the component, like document size, font, color, padding, and scrolling area. Now your question is, *what if want more than that?*

First things first, if you want to change few settings for specific section of your document, you can always do that with T_EX commands (e.g. *italics word* by `{\it italics word}`). *TEXDraw* has hundreds of them and we have covered them in separate document `reference.tex`.

Now, if you want to change properties for whole TEXDraw instances, or something that can't be configured just by T_EX commands alone, then this section is for you. Read thoughtfully because this section is slightly long-winded.

4.1 Adding or Removing Fonts

4.2 Activating SDF Font Rendering

4.3 Tweaking Default Configs

5 Extending *TEXDraw*

5.1 Adding Macros

5.2 Dealing with Sprites

5.3 Dealing with Links

5.4 Dealing with Images and other Media

5.5 Dealing with Interactive Scrolling

5.6 Dealing with Right-to-Left

5.7 TeX Input

6 Advanced Topics

6.1 How TEXDraw Works

6.2 Alignment Behaviour

6.3 Dealing Inputs from User

6.4 Extending TeX Supplements

6.5 Extending Macros at Code-Level

6.6 Extending Shaders

6.7 Upgrading prior to v5.0

7 About *TEXDraw*, Credits and Legal

7.1 About *TEXDraw*

7.2 BaKoMa Fonts 6

7.3 Credits